

The Architecture of Discrete Mechanics

Constraints, Cycle Surgery, and Macroscopic Equilibrium in a Change-First Algebra

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1. Introduction and Foundational Constraints

The architecture of discrete mechanics can be rigorously derived from an unusually austere foundation: a single operational mandate governing execution, formalized as the requirement that every executable state must possess a distinct legal successor. This principle, recognized as the continuability constraint (C_1), asserts that stasis is not merely unlikely, but fundamentally excluded from the dynamical domain. A state that maps to itself possesses no continuation, generates no difference, and accrues no history. Therefore, it is forbidden.

Before continuability can be enforced, the foundational gap across which change is measured must be established. This is defined structurally as the C_0 constraint, or the "comma" identity. The difference between a state that cannot change and one that can is the interstice that serializes a superposition into a readable distinction. A boundary walk on a cylinder yields no distinction and generates no time. Conversely, a walk on a Möbius strip reads both faces serially over successive steps, establishing the first bit of information and the genesis of temporal sequence.

1.1 The Difference Operator and the de Bruijn Shift

When the continuability constraint is coupled with a binary difference operator (Δ), the resulting dynamics are strictly constrained. Formalizing novelty as the count of distinct states visited up to a given time, a greedy novelty walk rigorously forces a maximal-length sequence without importing external linear-feedback shift register theory. For a linear difference recurrence over a finite field, the maximal-orbit choice corresponds to a characteristic polynomial generating an m -sequence, which successfully visits every nonzero state exactly once before returning.

However, the m -sequence contains a single fatal defect: the all-zero state maps to itself. Under a linear rule with no constant term, zero is a fixed point. This origin is the unique obstruction to C_1 in the entire construction, and its prohibition is absolute. The minimal repair to remove this fixed point is mathematically forced. To compel the zero state to change, its feedback bit must flip, compelling it to map to a specific nonzero state. This flip, however, fundamentally breaks bijectivity at the target state, as two distinct states now map to the same location.

To restore the bijection—a strict requirement for any reversible dynamics—the feedback of exactly one other state must also flip. Exhaustive computational search over all possible feedback modifications proves this repair is minimal (exactly size two), unique at every tested width, and exactly consists of the origin and its algebraic opposite. Within the class of difference recurrences, this repair is forced at every step, with no free choice remaining.

The repaired dynamics yield a single continuous orbit through every state, manifesting as a canonical de Bruijn shift. In this architecture, a uniform transition measure—where every legal transition appears exactly once—is not an assumed axiom of symmetry, but rather emergent exhaust. The universe of this algebra is the explicit resolution of a single collision between the structural mandate to change and the one mathematical state that structurally refuses to.

1.2 The Edge Lift and the Jet Tower

Analyzing the resultant dynamics as a walk on a graph where vertices represent states creates an artificial indeterminacy. If a vertex is reachable by multiple predecessors, asking what comes next is an ill-posed query, as the answer depends on the arrival history. This apparent branching is an artifact of measuring at the incorrect granularity. This is resolved by lifting the state from the vertex to the edge: the true state is not the current position, but the ordered pair of the previous position and the current one.

On the edge, evolution is a single-valued, strict bijection. The apparent fork at a convergent vertex completely disappears once the predecessor coordinate is restored. We confirm this on the balanced shift: every recurrent vertex is reached by exactly two predecessors, making the vertex-level successor genuinely two-valued. However, the edge-level evolution is a single cycle through all edges, and its inverse exists and composes back to the identity perfectly.

Consequently, there are exactly two maps in play. The first is the edge evolution, which is bijective and loses no information; this is the reader. The second is the projection from the edge back to the vertex, which permanently discards the predecessor coordinate. This projection is many-to-one and serves as the sole site of information erasure in the entire dynamical round. This perfectly mirrors the fold/hash split in cryptographic compression functions. For instance, the rounds of a standard hash compression function represent a bijective fold that preserves all information (recovered with zero failures by running the schedule backward), whereas the Davies-Meyer feed-forward addition is the non-injective hash where irreversible erasure occurs. Every application of the projection destroys exactly one bit of information at the convergent vertices where two histories meet, rendering the second law of thermodynamics a strict bookkeeping identity rather than a statistical tendency.

By lifting the state to the edge, derived mechanical observables organize into a successive jet tower. Position is the zeroth lift (J^0). The edge represents the first lift (J^1) and carries velocity. The triple of three consecutive positions represents the second lift (J^2) and carries acceleration. Empirical evaluation confirms that the second difference (Δ^2) is single-valued on edges but multi-valued on vertices. This establishes that mass and acceleration are not disparate phenomena requiring theoretical reconciliation, but are instead functionally native to the identical enriched edge state, representing successive reads of the same underlying trajectory.

2. The Bracket Algebra and Interaction Residues

Interaction within the transformation space is quantified not by scalar magnitudes, but by the commutator $= ABA^{-1}B^{-1}$, which serves as the primary measurement object. If the composite returns the identity, the interaction bill is zero. If it fails, the geometric residue encodes the exact physical trace of the interaction.

2.1 Direction, Torsion, and Curvature

The bracket algebra's operational table divides into specific commutation classes. Over a standard execution alphabet consisting of fundamental bijections (e.g., XOR, Addition, Rotation, Half-Swap, Multiplication, Negation), empirical evaluation reveals that the overwhelming majority of pairs do not commute. The analysis establishes that direction is not a posited axiom, but a derived property: a direction is defined strictly as a maximal commuting family.

The nature of the non-commuting pairs bifurcates precisely based on the presence or absence of a carry chain:

- **Torsion (Flat Twist):** Linear interactions lacking carry propagation, such as an XOR operation against a Half-Swap or a Rotation, consistently leave a constant numerical residue from every center evaluated. This represents a loop failing to close by an identical displacement everywhere, mapping to a uniform flat twist or torsion.
- **Curvature (Field):** When a carry chain enters the algebraic operation—such as Addition or Multiplication executed against another operator—the residue immediately becomes a highly variable field. For example, Addition against Rotation generates thousands of distinct residue values over an equal number of evaluated centers. Because the interaction price varies continuously with position, the carry chain is directly identified as the mathematical domain of geometric curvature. This structural property explains why cryptographic primitives relying on Addition-Rotation-XOR (ARX) natively mix state, effectively functioning as curvature farming within a purely bijective regime.

2.2 The Non-Closure of the Ledger

The bracket algebra definitively does not close. Iterating commutators multiplies the transformation vocabulary without re-minting library generators. Evaluated across three depths, the number of distinct transformations expands exponentially, with branching entropy rising steadily toward the theoretical free bound.

Because the residue is a state modification, the inheritance rules of the change-first algebra dictate that it must itself possess freedom and pay for subsequent motion. The interaction price re-enters the operational economy as a full generator. Consequently, time emerges purely as ledger depth—the cumulative bits of newly generated address space required to track the expanding interaction residue without resetting or looping into existing generators.

3. The Continuity Dichotomy: Mass versus Transparency

The commutator possesses two mathematically independent projections that correspond to fundamental physical properties. The first is the order of the commutator, representing the number of times the residue must be applied to close back to the identity. This settlement cost is mathematically identified as inertial mass: $O(\lambda) = \log_2 \text{ord}(\lambda)$. The second projection is the fixed-point count, representing the exact number of

states surviving the interaction unaltered, formalized as $F(\lambda) = \#\{i: L_i = 1\}$, which corresponds to the physical concept of transparency or visibility.

3.1 Refuting Displacement as Mass

Initial hypotheses within the discipline proposed that inertial mass represented displacement resistance, suggesting that a massive body should exhibit less absolute displacement under a uniform push. However, empirical measurement systematically refutes this. The displacement channel saturates immediately at the thermal floor (exactly $W/2$ bits) because every elementary generator acts as a full avalanche mixer. When measuring the displacement residue across varied bodies, the spread-to-mean ratio is an indistinguishable 0.008. In this metric, a lightweight structure and an inherently massive one scramble to identical thermal equilibria, providing absolutely no gradient for measurement.

Mass, therefore, must inherently reside in the settlement channel. By measuring the base-two logarithm of the order of the redirection element, the spread-to-mean ratio jumps by two orders of magnitude. Furthermore, because the mathematical order of a permutation is entirely independent of the starting point, mass calculated in this channel is center-independent. It represents an intrinsic, structural property of the body alone, completely decoupled from the relational accident of the state it happens to act upon.

3.2 The Uncoupling of Opacity and Inertia

A highly seductive early hypothesis posited that opacity (a lack of fixed points) and inertia (high order) were manifestations of the identical physical quantity, representing accumulated cross-term interaction history. The explicit separation of the commutator's independent projections decisively refutes this.

The decisive counterexample is a commutator that functions as a fixed-point-free involution. An involution has an order of exactly 2, yielding a settlement mass of exactly 1 bit—meaning the interaction is almost entirely free to reverse. Yet, because it is fixed-point-free, its transparency is precisely zero. It is simultaneously nearly weightless and completely opaque in the exact same interaction. Consequently, mass reads the commutator's global order, while opacity reads its local fixed points. The depth dial drives both metrics, but strict structural mathematics prohibits their equivalency.

3.3 The Formal Proof of the Continuity Dichotomy

The structural separation of mass and transparency is formalized by their profoundly irreconcilable continuity properties under geometric perturbation, establishing a strict dichotomy of fragility.

Theorem 4 (Fixed points are Lipschitz bounded): For any permutation $P \in S_\Omega$ and any generic surgery $\sigma \in S_\Omega$, the fixed point count satisfies a rigorous modulus of continuity:

$$||\text{Fix}(\sigma P) - \text{Fix}(P)|| \leq |\text{supp } \sigma|$$

Proof: Let $s = |\text{supp } \sigma|$ and let $U = \Omega \setminus \text{supp } \sigma$. For any $x \in U$ with $P(x) = x$, we have $(\sigma P)(x) = \sigma(x) = x$, establishing that $x \in \text{Fix}(\sigma P)$. Hence, $\text{Fix}(P) \cap U \subseteq \text{Fix}(\sigma P)$, which directly implies $|\text{Fix}(\sigma P)| \geq |\text{Fix}(P) \cap U| \geq |\text{Fix}(P)| - s$. Applying the identical symmetric argument to σ^{-1} and the permutation σP yields the inverse inequality, combining to form the exact Lipschitz bound. This establishes F as Lipschitz with a constant of 1 with respect to geometric support. A minimal support-3 surgery can only alter the fixed-point count by a maximum of 3, regardless of N , the current partition, or any other macroscopic feature.

Theorem 5 (The order has no modulus of continuity): For every $M > 0$, there exist an N , a permutation P on N points, and a transposition τ with $|\text{supp } \tau| = 2$, such that the settlement ratio $\text{ord}(\tau P)/\text{ord}(P) > M$. At

$N = 2^k$, the catastrophic ratio $N/4$ is definitively attained. *Proof:* Let $k \geq 4$ and $N = 2^k$. Let P be a single N -cycle, so $\lambda(P) = (N)$ and $\text{ord}(P) = N = 2^k$. Set the offset $s = N/2 - 1$. Because $k \geq 2$, $N/2$ is even, making s strictly odd. Choose $x \in \Omega$ and put $y = P^s(x)$. Let $\tau = (x, y)$. The cycle splits into two cycles of lengths s and $N - s$. The new order is $\text{lcm}(s, N - s)$. Because s is odd, any common divisor of s and $N - s$ must divide their sum $N = 2^k$, which is a power of 2. The only such divisor is 1, establishing $\gcd(s, N - s) = 1$. Therefore, the new order is $s(N - s) = (N/2 - 1)(N/2 + 1) = N^2/4 - 1$. The ratio of new to old order is $(N^2/4 - 1)/N = N/4 - 1/N$, which is entirely unbounded over transpositions of fixed support 2.

The profound asymmetry between these two channels is structurally derived from the exponent map:

$$O(\lambda) = \sum_q e_q(\lambda) \cdot \log_2 q$$

where $e_q(\lambda) = \max_i v_q(L_i)$ is the maximum q -adic valuation across all cycle lengths. The order is a sum over prime maxima. A mathematical maximum inherently lacks locality; a single newly minted prime contributes its full weight regardless of its spatial location within the partition spectrum. In contrast, F is a localized geometric count. The continuity dichotomy is therefore an absolute mathematical consequence of the channel definitions, not a byproduct of the transition dynamics.

4. Formal Theory of Permutation Surgery

The mechanism actively driving the spectral evolution is not geometric perturbation support, but precise cycle topology. Holding a perturbation size fixed at exactly 3 points yields wildly divergent changes in the order channel depending on how those points span the existing cycles. A span-1 surgery moves the order by under 4 bits; a span-3 surgery moves it by 5 bits; and a span-2 surgery moves it by over 9 bits, representing a factor exceeding 2.4 at constant geometric support. The exact local surgery rules dictate the arithmetic outcome.

4.1 The Transposition and Factorization Rules

Theorem 1 (Transposition Surgery): Let $P \in S_\Omega$, let $x \neq y$ in Ω , and let $\tau = (x, y)$. The cycle structure $\text{Cyc}(\tau P)$ agrees with $\text{Cyc}(P)$ everywhere outside the specific cycles containing x and y , subject strictly to two rules :

1. **Split:** If x and y lie in a common cycle C of length L , and $y = P^s(x)$ with $1 \leq s \leq L - 1$, then C is deterministically replaced in $\text{Cyc}(\tau P)$ by two distinct cycles of lengths s and $L - s$. Explicitly, the split partitions the cycle into an arc beginning at x of length s , and an arc beginning at y of length $L - s$.
2. **Merge:** If x lies in cycle C_1 of length L_1 and y in a distinct cycle C_2 of length L_2 , they merge into a single cycle of length $L_1 + L_2$. The resulting sequence visits every point of C_1 followed sequentially by every point of C_2 . A transposition reliably changes the total cycle count by exactly ± 1 .

Theorem 2 (Factorization): For any distinct points $a, b, c \in \Omega$, the 3-cycle factors algebraically as $(a, b, c) = (a, c)(a, b)$. Consequently, the precise effect of any minimal-support 3-cycle surgery is completely resolved by applying Theorem 1 twice sequentially.

Corollary 2.1 (Span Rules): The sequential application dictates the cycle count change (ΔK) based on the topological span $\sigma \in \{1, 2, 3\}$:

- **Span 3** ($\sigma = 3$): All three points occupy distinct cycles. Two sequential merges occur, yielding $\Delta K = -2$. The three original cycles merge into one massive cycle of length $L_a + L_b + L_c$.
- **Span 2** ($\sigma = 2$): Two points share a cycle, the third does not. One split and one merge must occur sequentially. Regardless of the internal ordering, the cycle count remains perfectly static, yielding $\Delta K = 0$.
- **Span 1** ($\sigma = 1$): All three points share a single cycle. The first application splits it. Whether the second application merges or splits again depends entirely on the cyclic orientation of a, b, c along the cycle relative to the surgery's innate orientation. This yields $\Delta K \in \{0, +2\}$.

4.2 The Compression Law

Theorem 3 (Compression Law): The successor spectrum is computable in $O(L_{\max})$ time (where $L_{\max}$ is the largest affected cycle length) entirely from localized data, without ever executing permutation matrix composition.

The transition algorithm requires only a triple of localized index-position pairs: $((i_a, \pi_a), (i_b, \pi_b), (i_c, \pi_c))$. It applies the transposition rule twice. For a split, it rotates the stored internal sequence so the targeted point leads, and splices the array at the calculated offset s . For a merge, it rotates both target arrays to their respective points and concatenates them.

This compression law was verified against absolute ground truth obtained by explicit full-matrix composition and decomposition. Trials were executed across lane widths of $W \in \{8, 10, 12\}$, utilizing both pure bases (multiplications where all cycle lengths are powers of 2) and rich bases (uniform random permutations). Over 1,800 executed trials distributed exhaustively across all three span classes, the predictor yielded the exact successor spectrum in 1,800 out of 1,800 cases, cementing the transition as completely closed-form.

5. The Arithmetic Mechanism and Spectral Fragility

The underlying thermodynamic driver of the entire mechanics is formally stated in Theorem 6: prime injection via the closure failure of the prime-power set under integer addition.

Theorem 6 (Prime injection by merge): Let a permutation P have all cycle lengths equal to powers of a single prime π , rendering its prime support $S(\lambda(P)) \subseteq \{\pi\}$. Let a surgery merge two cycles of lengths π^a and π^b with $a \geq b$. The merged cycle attains a length of:

$$\pi^a + \pi^b = \pi^b(1 + \pi^{a-b})$$

If $a > b$, the expression $1 + \pi^{a-b} > 1$ and is guaranteed to possess at least one prime divisor. Any divisor of $1 + \pi^{a-b}$ is strictly coprime to π . Therefore, the merge strictly and inevitably enlarges the prime support, injecting new primes into the arithmetic baseline.

This theorem provides an elegant arithmetic explanation for the profound fragility of ordered objects. An ordered, coherent body built entirely from a commuting family possesses a mathematically pure spectrum. For example, multiplications by odd constants generate cycle lengths that are entirely powers of 2, possessing exactly one distinct prime. Because the spectrum is pure, any topological merge forced by a surgery inevitably throws the resultant length out of the prime-power set. Merging cycles of lengths 4, 8, and 16 produces a cycle of length 28, which factors as 4×7 , injecting the prime 7 into a purely 2-adic spectrum. Merging 1024, 512, and 128 yields 1664, factoring as 128×13 , immediately injecting the prime 13.

Conversely, a highly thermalized, rich spectrum (such as a uniform random permutation containing 8 or 9 distinct primes) is robust. Any newly formed sums are highly likely to factor into primes already present within the vast existing support $S(\lambda)$, leaving the least common multiple entirely unmoved.

Empirical validation confirms this absolutely. Identical 3-point surgeries applied to pure spectra ($W = 12, |S| = 1$) generated massive mean order gains of +6.20 and +6.35 bits. The identical surgeries applied to rich spectra ($|S| = 8, 9$) generated changes of -2.56 and -0.01 bits.

This firmly establishes that fragility and structural coherence are identical mathematical properties read through two different lenses. The apparent step-function collapse of coherence, the doubling of mass under a perturbation of less than a tenth of a percent, and the specific sensitivity of the settlement channel all reduce to this arithmetic fact: coherent bodies live on arithmetically pure spectra, and pure spectra possess nowhere to go but up when topologically perturbed.

6. The Fragmentation Theorem and Settlement Scaling

The settlement mass accumulates with the depth of the body up to an absolute ceiling defined by the configurational capacity of the finite state space. However, the trajectory of that scaling depends entirely on one specific algebraic property of the interaction: whether it possesses a defined direction.

Generic transformations—those that differ mathematically from their own inverse—represent directed circulation. Under directed pushes, the settlement order scales along the generic line governed by the Erdős–Turán ceiling, characterized by $O(\log^2 N)$ scaling.

Conversely, an involution is a transformation equal to its own inverse, representing directionless reciprocation. When the push is an involution, the dynamics instantly drop out of the generic regime and adhere to a strict square-root scaling line, $O(\sqrt{N})$. Direction, therefore, serves as the fundamental selector between two completely divergent asymptotic laws.

This phase transition is structurally forced by the **Fragmentation Theorem**. The canonical involution is the half-swap, a transformation that exchanges the two halves of a binary word. The exact fixed-point set of the half-swap constitutes the diagonal of the space (where the high half equals the low half), sized exactly $\sqrt{N}$. The commutator of any structural body with this involution analytically expands to a product of two involutions, each fixing exactly $\sqrt{N}$ points.

Extensive numerical validation confirms the governing fragmentation mechanic: a product of two involutions, each fixing k points, inherently fragments the space into approximately k distinct cycles, with a mean cycle length scaling strictly to N/k . When $k = \sqrt{N}$, the geometry of the state space enforces exactly $\sqrt{N}$ cycles of length $\sqrt{N}$. This massive structural fragmentation occurs *before* any least-common-multiple arithmetic is computed, forcibly mapping the settlement order to the square-root scale.

The coefficient C of this scaling line, defined as $C = \log_2(\text{order})/\sqrt{N}$, was initially hypothesized to be a universal constant. However, rigorous cross-ensemble controls decisively refuted universality. Measured across space sizes spanning a sixty-four-fold range, the coefficient sits between 1.67 and 1.69 for uniformly random involution pairs, while under-mixed shallow bodies sit noticeably lower at 1.45. The $O(\sqrt{N})$ scaling itself is a rigidly proven consequence of geometric fragmentation, while the coefficient C is properly categorized as an empirical calibration strictly dependent on ensemble mixing depth.

7. The State Transition as a Markov Chain

7.1 Sufficiency of the Cycle Type

Under a label-independent, uniform surgery selection protocol, the exact permutation matrix carries absolutely no privileged information beyond its cycle spectrum.

Theorem 7 (Sufficiency of the cycle type): Any two permutations P and P' within S_Ω possessing identical cycle types are conjugate, meaning there exists a $g \in S_\Omega$ such that $P' = gPg^{-1}$. Conjugation acts bijectively on the set of ordered triples of distinct points. Therefore, pushing the uniform measure forward through the standard conjugation formula guarantees that the induced probability distribution on successor spectra depends solely on the current partition.

To empirically validate this proof, the transition law of a pure multiplication was statistically compared against a randomly relabeled conjugate. The total variation distance between the distributions was measured at 0.2442. A critical same-source control, comparing two completely independent 4,000-surgery samples drawn from the exact same permutation, yielded a total variation distance of 0.2398. The mathematical discrepancy is exclusively the noise floor produced by a vast support. The cycle partition is a mathematically sufficient statistic, effectively reducing the entire complex dynamics to a formal Markov chain defined on integer partitions of N (Λ_N).

7.2 The High-Entropy Trajectory

Despite possessing a perfectly closed-form microscopic transition algorithm (Theorem 3), the trajectory of the partition definitively does not close. From a single isolated starting partition at a lane width of $W = 10$, 4,000 uniform random surgeries violently branched into 972 distinct successor spectra. The most likely individual successor captured only 0.104 of the probability mass, yielding an empirical conditional entropy $H(\lambda_{t+1} | \lambda_t) \geq 7.49$ bits per step. Because the vast support size exceeds the empirical sample size, this serves strictly as a lower bound. The profound complexity of this dynamical mechanics resides wholly in the high-entropy orbit of the partition under sequential surgery, rendering it immune to any low-dimensional geometric invariants.

8. The Macroscopic Equilibrium and Drift Field

While the microscopic trajectory branches relentlessly, a specific scalar projection of that trajectory yields a strict, deterministic macroscopic equilibrium. This bridges the gap between chaotic microscopic interaction and predictable macro-scale law.

Defining the drift field $D(p)$ as the expected change in order given the current prime count p :

$D(p) = \mathbb{E}[O(\lambda_{t+1}) - O(\lambda_t) | p(\lambda_t) = p]$ The step change cleanly decomposes into two antagonistic thermodynamic forces, defined by Equation 0.1 [1]: $\Delta O = G - \Lambda$

Here, G is the gain from newly minted prime factors injected by merges, and Λ is the loss from eroded prime-power exponents degraded by topological cycle splits.

Empirical trajectory mapping, measured over 2,000 sequential steps from four distinct starting spectra at $W = 12$, demonstrates that G remains remarkably flat regardless of the existing prime count p . Conversely, Λ scales linearly with p , possessing a slope of approximately 1.3 bits per prime.

p	n	E[G]	E[Λ]	D(p)
5	91	11.15	6.46	+5.58
6	220	11.38	9.67	+2.52
7	394	12.09	10.73	+2.10
8	483	11.50	12.08	+0.18
9	423	11.16	12.81	-0.92
10	235	9.75	14.19	-3.71
11	98	8.37	15.21	-6.02
12	29	8.83	15.71	-6.29

The drift field falls monotonically, crossing zero between six and seven primes. Linear interpolation precisely locates this stable fixed point at $p^* \approx 6.2$ distinct primes for $W = 10$. Below this equilibrium, the prime injection rate dominates and order grows; above it, exponent erosion dominates and order decays.

Extensive convergence testing verified this stability. Trajectories of 1,200 steps were launched from drastically disparate initial conditions: a pure 1-prime spectrum ($|S| = 1$), a rich 5-prime random spectrum, and an extremal spectrum consisting of one giant N -cycle. All three trajectories inevitably terminated in the exact same arithmetic neighborhood, fluctuating tightly between 5 and 10 primes with orders settling in the twenties to forties of bits.

The zero of the drift field obeys an empirical width-dependent scaling law :

W	N	E[G]	c	p* predicted	p* measured
8	256	5.67	0.870	4.11	4.71
10	1024	8.02	1.108	5.71	6.22
12	4096	11.35	0.974	7.54	8.12

W	N	E[G]	c	p* predicted	p* measured
14	16384	14.24	0.583	8.74	10.32

A least-squares fit confirms the scaling: $p^*(W) \approx 0.94 \cdot W - 2.8$. This establishes a macroscopic statistical-mechanics paradigm generated entirely from a software constraint.

9. The Flatness of the Observable Layer

Within computational mechanics, observables function dynamically as active intervention ports. They passively read a specific functional of the state and feed it back into the execution loop to constrain the subsequent selection.

9.1 Single-Port Attractors

Subjecting the Markov chain to single-observable feedback definitively proves coordinate independence. Over 150 steps from a fixed origin ($W = 10$), steering the chain using distinct ports yields highly divergent structural attractors :

port	O (bits)	F	K	M	p
start	8.00	2.00	19.00	256.00	1.00
minimize O	10.08	2.20	11.80	944.80	2.20
maximize F	28.89	4.20	13.80	660.60	7.00
minimize K	11.92	1.00	2.60	945.40	2.60
minimize M	40.79	1.40	13.40	232.80	9.40

Every port mathematically optimizes its own specific coordinate while actively destroying the others. The loop maximizing fixed points suffers from nearly the worst order (28.89 bits). The loop minimizing the largest cycle triggers a catastrophic mass explosion, pushing the order past 40 bits and scattering the prime count to 9.40.

The dimensional asymmetry here is immense. A 1024-point state space requires exactly $\log_2(1024!) \approx 8769$ bits to fully describe. Yet, the cycle-count port (K) feeds back fewer than 2 bits of information per iteration. A microscopic pipeline constraint reliably steers a massive configuration state space into designated macroscopic attractor basins.

9.2 Refuting the Compositional Algebra Hypothesis

A prevailing hypothesis suggested that observables might compose into a higher-order constraint algebra—that feeding two orthogonal ports back jointly would force the system into a novel, emergent attractor class unattainable by either port individually.

An initial test seemingly confirmed this non-additivity, but was subsequently invalidated due to a structurally infeasible baseline. The naive additive prediction added individual coordinate displacements, yielding mathematically impossible physical states (e.g., generating negative cycle counts of -9 and max lengths of 1768 within an $N = 1024$ space). Because the state space of integer partitions is strictly constrained ($\sum L_i = N, K \geq 1, M \leq N$), any additive deviation from an impossible baseline registered identically as a false positive departure.

A rigorous, corrected geometric test evaluated whether the two-port joint attractor resided linearly on the spatial segment directly interpolating the two single-port attractors. The normalized distances from the interpolative segment ($d/|seg|$) were measured across 5 random seeds :

pair	type	d/ seg
$O \downarrow$ with $F \uparrow$	arithmetic $\times$ geometric (local)	0.11
$O \downarrow$ with $K \downarrow$	arithmetic $\times$ geometric (global)	0.29
$p \downarrow$ with $M \downarrow$	arithmetic $\times$ scale	0.20
$O \downarrow$ with $p \downarrow$	arithmetic $\times$ arithmetic (control)	0.35
$F \uparrow$ with $K \downarrow$	geometric $\times$ geometric (control)	0.25

Every pairing landed mathematically on the segment. The largest departure actually belonged to the arithmetic-by-arithmetic control pair, utterly inverting the registered prediction.

The hypothesized compositional algebra is definitively refuted. The observable layer operates exclusively as a flat coordinate system. Joint feedback merely forces a weighted Euclidean compromise between singular optima.

10. Numerical Protocols and Exact Verifications

To satisfy analytical rigor, the experimental protocols utilize exact integer arithmetic exclusively, completely avoiding floating-point heuristics for structural decisions. Prime factorizations are obtained via strict smallest-prime-factor sieves, and mathematical orders are calculated by prime-power accumulation over specific cycle lengths rather than direct least common multiple computations, mathematically averting overflow limits on immense orders.

Three foundational families are systematically utilized: Pure forms (multiplications by an odd constant modulo 2^W , yielding exactly $|S| = 1$), Rich forms (uniform random permutations yielding $|S| \in \{8,9\}$), and Extremal forms (single N -cycles). Evaluated lane widths spanned $W \in \{6,8,10,12,14\}$.

Continuity measurements swept cross-generator support from 0 to 48 points on a space of 4096. At $W = 12$, holding the scaffold fixed, the order skyrocketed predictably from $10.00 \rightarrow 19.65 \rightarrow 23.50 \rightarrow 28.65 \rightarrow 34.55 \rightarrow 37.46$ bits, while the fixed-point count remained mathematically frozen at exactly 4.00 through that entire domain, definitively proving the continuity dichotomy.

For span classification, 3,000 minimal-support 3-cycle surgeries generated strict compliance: 1,765 span-3 cases produced $\Delta K = -2$ with 100% precision; 1,140 span-2 cases produced $\Delta K = 0$ with 100% precision; and the merge-length law (sum of lengths) held perfectly in 231 out of 231 qualifying trials. The compression law (Theorem 3) maintained exact state-spectral predictive accuracy over 1,800 out of 1,800 random trials.

11. The Ledger of Refuted Claims and Epistemological Boundaries

To maintain strict architectural boundaries, hypotheses that failed empirical execution are meticulously logged with their balancing structural expansions. The framework evolves by narrowing its foundation, utilizing refutation as a mathematical tool.

- **R1: Mass is displacement resistance.** Falsified. Every elementary band transformation avalanches. The Hamming channel saturates immediately at exactly $W/2$ bits with no measurable gradient. The expansion relocated mass permanently to the settlement channel—the mathematical order of the commutator—providing a measurable, center-independent intrinsic metric.
- **R2: Opacity is inertia.** Falsified. A fixed-point-free involution commutator has minimal mass (exactly order 2, approx. 1 bit) but total opacity (exactly 0 fixed points). Mass and transparency are structurally independent functional projections of the identical underlying commutator.
- **R3: Maximum structural persistence is $\sqrt{N}$.** Falsified. While dynamic law cannot fix any points, objects inside the algebra can fix up to $N - 3$ states. The mathematical quantum of change is absolutely 3, firmly decoupling object persistence from dynamic execution constraints.
- **R4: Coherence is binary.** Falsified. A tunable cross-generator demonstrates continuous, massive degradation in the settlement order channel based on commutation deficit. Fragility is a property of the specific measurement channel (e.g., LCM), not an inherent structural state of the body.
- **R5: Perturbation support drives the response.** Falsified. Fixing geometric support at precisely 3 points while altering the topological span yields order responses differing by a factor of over 2.4. Cycle topology and arithmetic prime injection govern the outcome, absolutely not geometric support.
- **R6: Observables compose as an algebra.** Falsified. Corrected geometric Euclidean baselines prove the observable layer is structurally flat and non-compositional.
- **R7: p^* is a universal constant.** Falsified. The drift equilibrium is highly width-dependent, scaling linearly as $p^*(W) \approx 0.94W - 2.8$.
- **R8: External quadratic form mapping to lepton masses.** Audited and rejected as egregious overfit. The proposed form $F(p, q) = \alpha p^2 - \alpha pq + q^2$ utilized a free parameter $\alpha = 4.08222$, sitting within one part in a thousand of the value (49/12) that annihilates the reference denominator. This constitutes an arithmetic near-cancellation. The target ratios require α to six significant figures, with a minimal shift of 10^{-5} producing catastrophic 10% errors. The sole parameter-free limit missed its targets by 0.55%, and an exhaustive search yielded 392 different

coprime integer triples satisfying the constraints to within 0.5%. External physical mappings devoid of internal mathematical derivation are explicitly excluded from the theoretical architecture.

- **Rg: Stated bound reported as value.** Corrected. The state information at $N = 1024$ was originally reported using a truncated factorial guard yielding roughly 1,019 bits. The mathematically correct exact calculation is $\log_2(1024!) \approx 8769$ bits. Correcting this error widens the gap between the control port and the configuration state by a factor of 8, massively strengthening the architectural argument regarding port bandwidth asymmetry.

12. Conclusion

The discrete mechanics derived from the continuability constraint (C_1) and the difference operator ultimately isolate complexity into a specific, mathematically defined zone. The surviving architecture relies on three precise layers. First, the Verb Layer acts entirely via executable operations, specifically minimal-support 3-cycle surgeries. Second, the State Layer exists strictly as the evolving integer cycle partition. The microscopic state transition is entirely closed-form, dictated by rigorous topological split and merge rules operating independent of heavy matrix composition. Because the cycle type is a mathematically sufficient statistic, the system manifests uniquely as a high-entropy Markov chain branching at > 7.49 bits per step on integer partitions.

Finally, the Noun Layer operates as flat, non-evolving mathematical projections of the state, including inertial mass (via prime exponents) and transparency (via length-1 multiplicity). These observables serve as highly efficient constraint ports for dynamical feedback, successfully steering thousands of bits of state using fewer than 2 bits of control, but possess no internal compositional algebra of their own.

The complexity of the system is neither in the flat nouns, the trivial ports, nor the closed-form local transition rules. The irreducible complexity lives entirely within the broad branching orbit of the partition under repeated surgery. While the microstate does not close, the macrostate yields a definitive arithmetic equilibrium ($p^*(W)$) governed exclusively by the prime injection mechanisms of integer addition. The mathematical framework is completely secure; its physical interpretation remains strictly downstream of its rigorous computational proofs.